

RESEARCH QUESTION

Can machine learning catch more attacks without drowning us in false alarms?

We compare four machine-learning models on the same reproducible network-flow study and judge them on two things anyone can understand: attacks found and false alarms created.

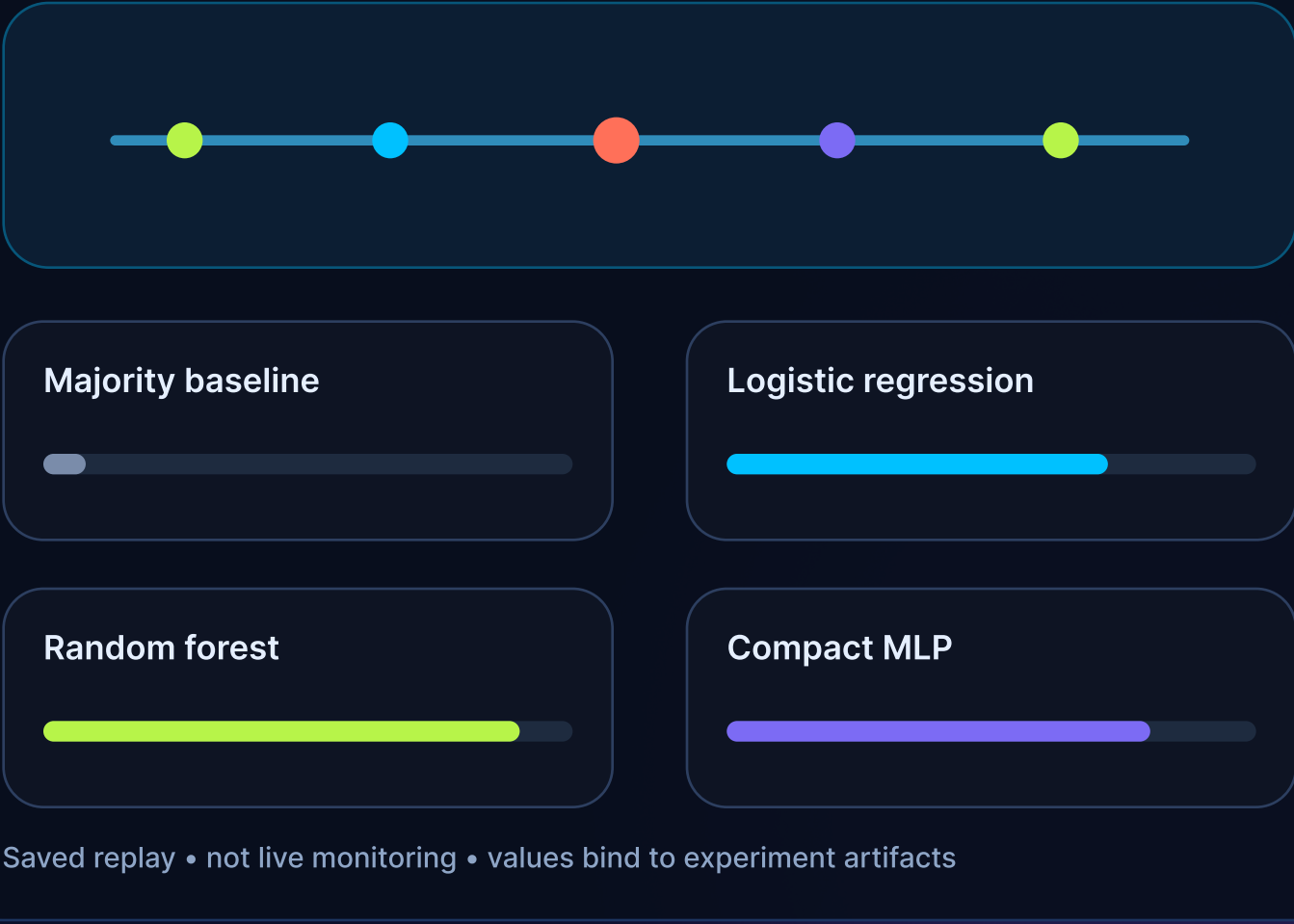
Watch the demo

See the benchmark

VISUAL PROJECT DEMO

One saved flow → four model decisions

The demo replays evidence already produced by the experiment.



STUDY AT A GLANCE

4
Models compared

12,000
Valid synthetic flows

5
Traffic periods

≤10%
False-alarm cap

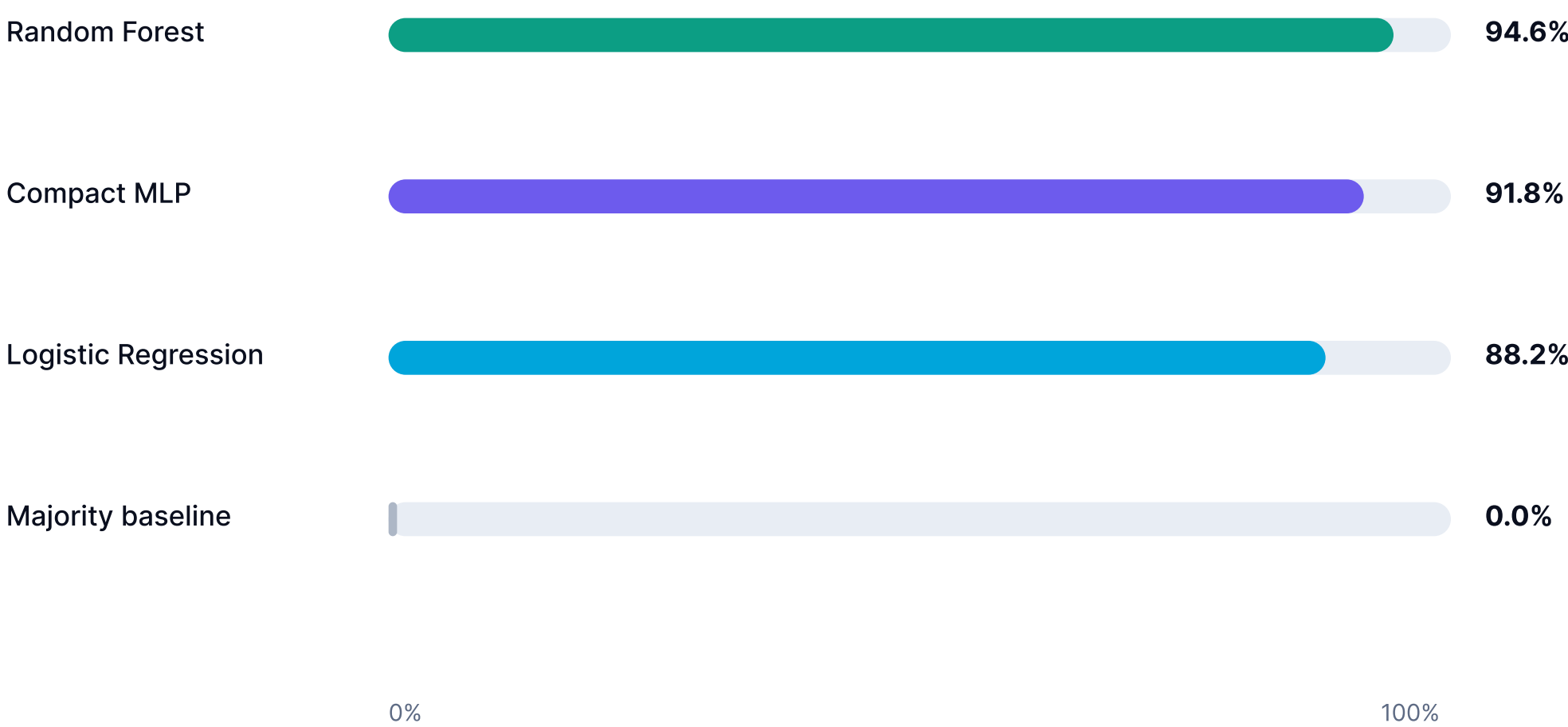
BENCHMARK

The benchmark tells the story at a glance.

Example saved-evidence values from the current project UI brief. The implemented website should bind these views to metrics.csv.

Attack detection

Higher is better — how many attack rows the model finds



RESEARCH ANSWER

Which model gives the best balance?

Random Forest

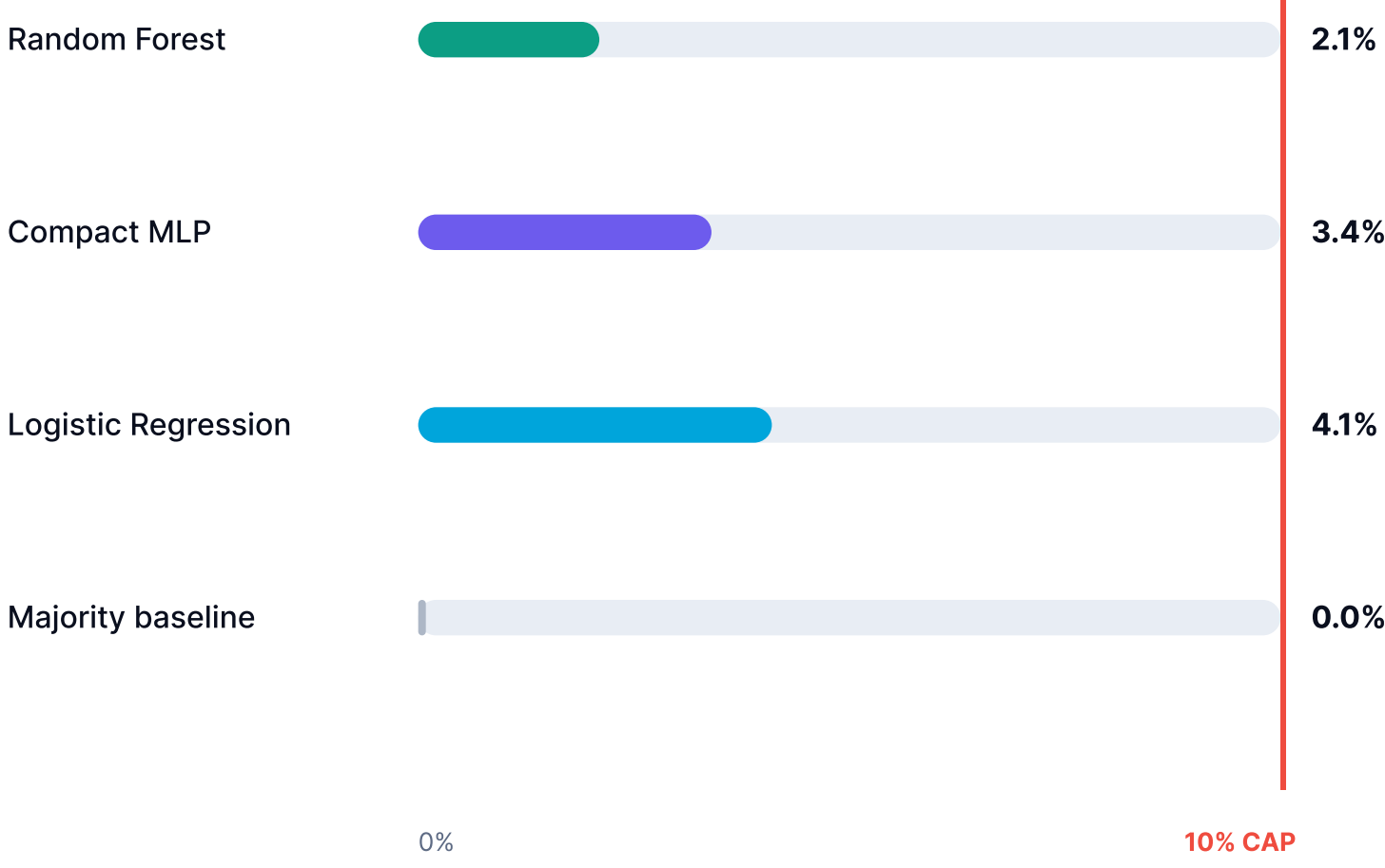
In the current saved-evidence example, it finds the most attacks while remaining well below the study's 10% false-alarm limit.

94.6% detection • 2.1% false alarms

Research rule: maximize attack detection among models staying at or below 10% FPR.

False alarms

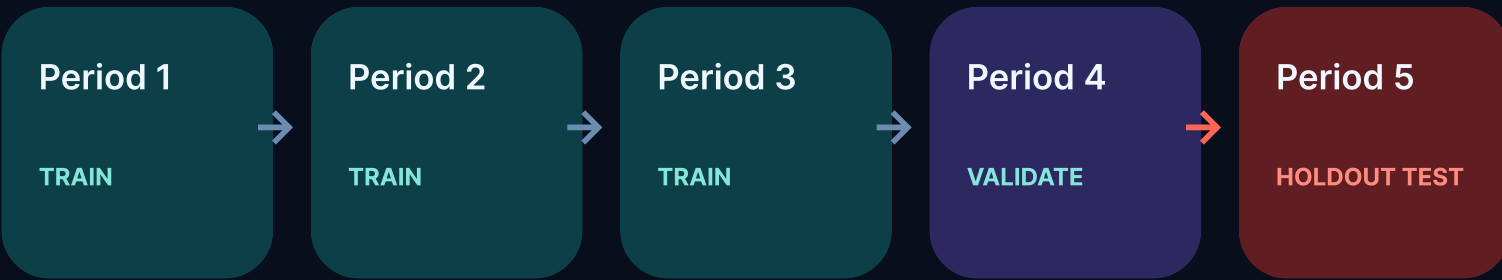
Lower is better — benign traffic incorrectly flagged as attack



WHY TEST ON LATER TRAFFIC?

Traffic changes. The test should too.

Instead of only shuffling the same data, the temporal split trains on earlier periods and checks whether the detector still works on later traffic.



Research question

Does the model still work when later traffic patterns change?

RESEARCH QUESTIONS → ANSWERS

The project should answer questions, not just show metrics.

01

Which model wins?

Random Forest in the current saved-evidence example. It gives the strongest attack detection while staying comfortably below the false-alarm cap.

02

Why not use accuracy?

A detector can look “accurate” while still missing attacks or constantly flagging benign traffic. Detection and false alarms tell a clearer story.

03

Why temporal testing?

Because later traffic can look different. The temporal holdout asks whether the model still works when patterns shift over time.

04

Why synthetic data?

The same declared seed recreates the same study, keeping the benchmark self-contained, reproducible, and easy to audit.

PROJECT DEMONSTRATION

Watch one saved flow pass through four models.

Switch between an attack-like replay and a benign replay. Figma Smart Animate changes the packet pattern and each model's saved probability bar.

Open demo →

ATTACK-LIKE SAVED FLOW



BOTTOM LINE

The useful model is not the one with the prettiest accuracy score — it is the one that finds attacks while controlling false alarms.

Synthetic benchmark • binary BENIGN vs ATTACK • reproducible research

PROJECT DEMONSTRATION

Watch one saved network flow pass through four models.

Switch between an attack-like replay and a benign replay. The page explains what the models saw and how their saved scores changed.

Attack-like example Benign example

ATTACK-LIKE SAVED FLOW

Traffic signature

The replay highlights a few intuitive flow characteristics. These are descriptive cues for the visual, not live packet capture.



- High packet rate
- High SYN activity
- Shorter duration
- Payload anomaly

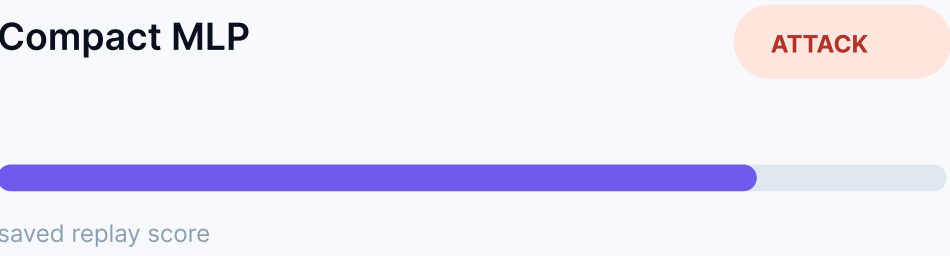
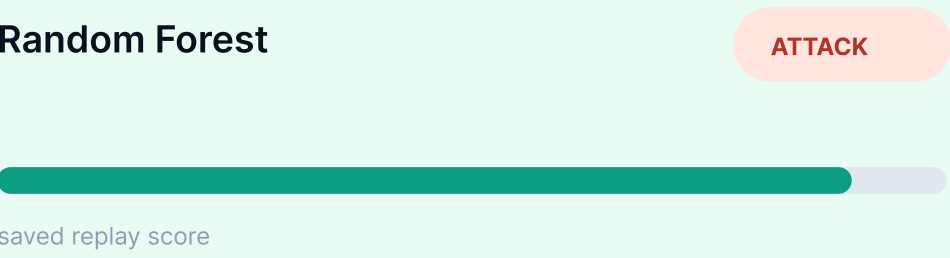
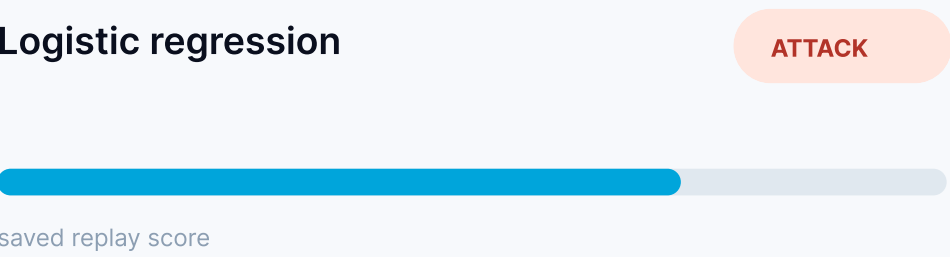


WHAT THIS DEMO MEANS

This is a deterministic saved replay. No live network is monitored, no model is retrained, and no new traffic is scored here.

Model decisions

Each bar is a saved replay score. Exact values come from the implemented demo artifacts.



PROJECT DEMONSTRATION

Watch one saved network flow pass through four models.

Switch between an attack-like replay and a benign replay. The page explains what the models saw and how their saved scores changed.

Attack-like example

Benign example

BENIGN SAVED FLOW

Traffic signature

A calmer saved flow pattern from the replay. The same four models receive the same prepared feature format.



Steady packet rate

Balanced flag activity

Longer duration

Stable byte pattern

01

Prepare

Clean + scale



02

Score

Run 4 models



03

Compare

Attack vs benign



04

Review

Explain result

WHAT THIS DEMO MEANS

This state replays a saved benign example so visitors can compare how the same four models respond to calmer traffic.

Model decisions

Each bar is a saved replay score. Exact values come from the implemented demo artifacts.

Majority baseline

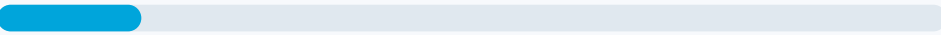
BENIGN



saved replay score

Logistic regression

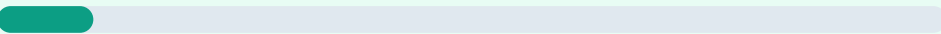
BENIGN



saved replay score

Random Forest

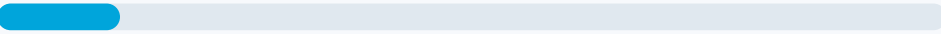
BENIGN



saved replay score

Compact MLP

BENIGN



saved replay score

RESEARCH ANSWER

Which model gives the best balance?

The study does not reward “accuracy” alone. A useful detector should find attacks while keeping false alarms under control.

ANSWER FROM SAVED EVIDENCE

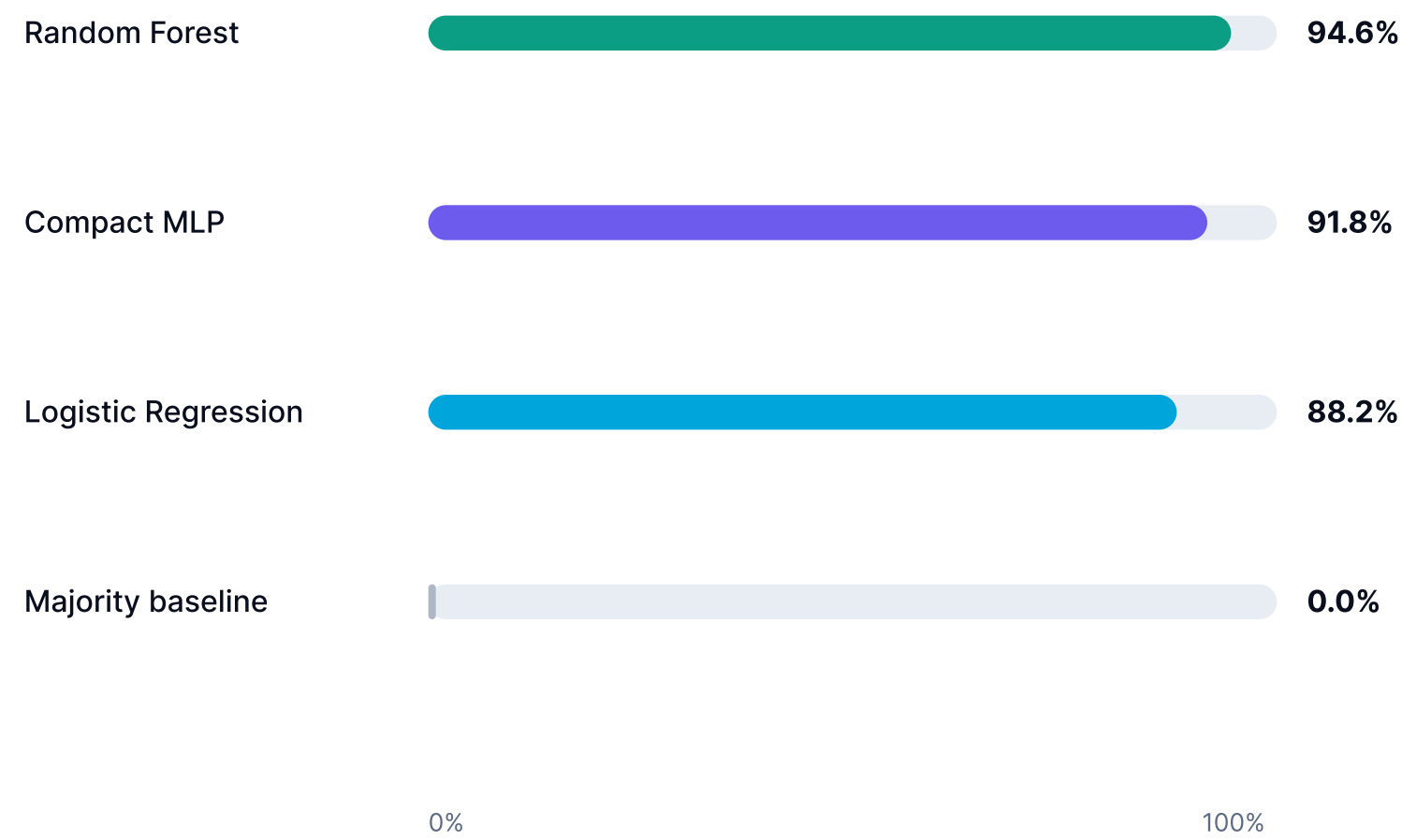
Random Forest

Strongest attack detection in the current saved-evidence example while remaining comfortably below the 10% false-alarm cap.

Maximize detection among models with $FPR \leq 10\%$.

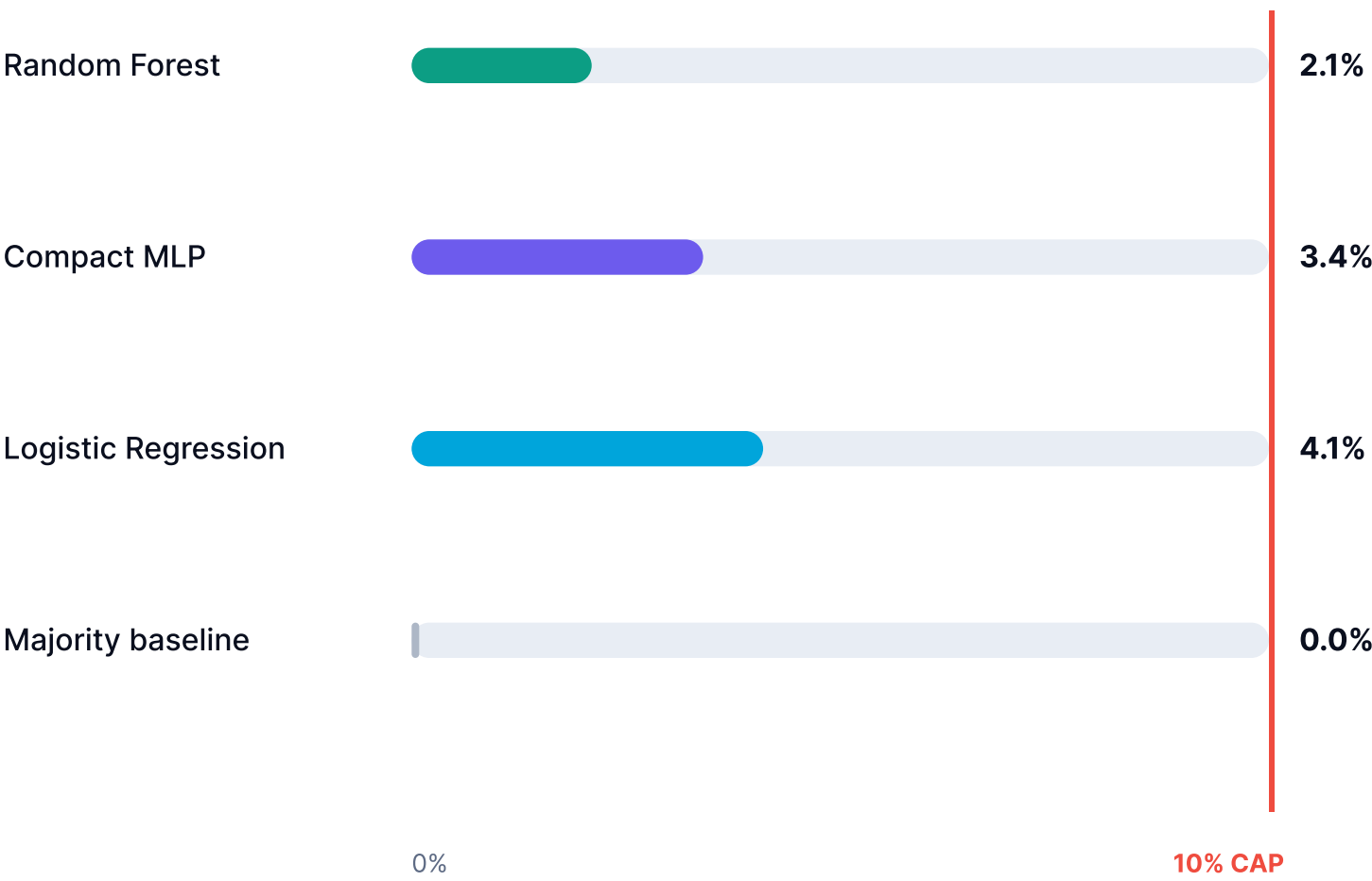
Attack detection

Higher is better — how many attack rows the model finds



False alarms

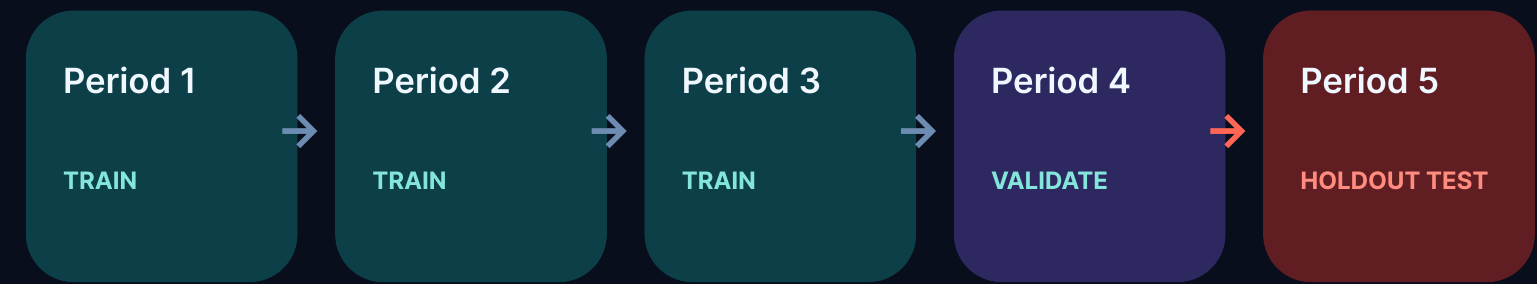
Lower is better — benign traffic incorrectly flagged as attack



WHY TEST ON LATER TRAFFIC?

Traffic changes. The test should too.

Instead of only shuffling the same data, the temporal split trains on earlier periods and checks whether the detector still works on later traffic.



RESEARCH QUESTION

Does the detector still work when later traffic changes?

Earlier periods teach the model. Period 4 is used to validate. Period 5 stays untouched until final holdout evaluation.

Research question → answer

Q Which model wins?

A Random Forest in the current saved-evidence example.

Q Why cap false alarms?

A Because high recall is not useful if benign traffic is constantly flagged.

Q Why temporal testing?

A It checks whether the detector still works when later traffic patterns change.

Q Why synthetic data?

A The same declared seed recreates the same study, keeping it reproducible and auditable.